



FADED'S THE HIVE

Comprehensive Server Guide

A cleaned-up, operator-friendly reference for running The Hive mod suite on Project Zomboid Build 42 servers.

GAME	Project Zomboid - Build 42
MOD SUITE	The Hive • Faded Trade Network • Faded Afflictions • CSR Claims
SIDE	Server-authoritative - all logic runs server-side
DATE	2026

[OPERATOR NOTE] Built for community server operators: quick tuning, readable tables, and clean reference sections without the old cluttered layout.



Operator Quick Start

Use this first when setting up or tuning a server. The full reference begins after the table of contents.

Start with defaults

For normal servers, use the Standard Survival profile first. Tune difficulty only after one real test night.

Tune pressure first

RaidsPerNight, NightCallRadius, AlphaPercent, and NightHordesPerNight are the biggest feel changes.

Protect performance

For larger servers, reduce MigrationRingTiles and RaidRecruitTiles, lower horde counts, and enable RenderCap.

Avoid conflicts

The Hive overrides zombie speed and behavior. Do not stack it with other zombie speed/AI overhaul mods unless tested.

[WARNING] Big warning for new players: Night Call, Raids, climbers, reanimation, and toxicity can stack. Do not max everything at once unless your group wants chaos.

Best places to start reading

- Section 18: Complete Sandbox Settings Reference - every setting and default value.
- Section 19: Recommended Server Configurations - casual, standard, hardcore, and large-server profiles.
- Section 24: Performance Considerations - safest dials for reducing lag.
- Section 25: Troubleshooting & Debug Tools - common problems and what to check first.



Table of Contents

The PDF also includes bookmark entries for fast navigation in most PDF viewers.

01	1. Overview & Design Philosophy
02	2. Architecture - How The Hive Is Built
03	3. Core System: Shared Mind (Pheromone Layer)
04	4. Core System: Per-Zombie Behavior
05	5. Core System: Alpha Leaders
06	6. Core System: Raid System
07	7. Core System: Night Hordes
08	8. Core System: Climb System
09	9. Core System: Mutation System
10	10. Core System: Reanimate System
11	11. Core System: Population & Repopulation
12	12. Core System: Dark Night
13	13. Core System: Night Call
14	14. Core System: Tactics & Feral Marking
15	15. Core System: Toxicity
16	16. Core System: Migration
17	17. Timing & Phase Reference
18	18. Complete Sandbox Settings Reference
19	19. Recommended Server Configurations
20	20. Companion Mod: Faded Trade Network
21	21. Companion Mod: Faded Afflictions
22	22. Companion Mod: CSR Claims System
23	23. Data Persistence & Save/Load
24	24. Performance Considerations
25	25. Troubleshooting & Debug Tools



1. Overview & Design Philosophy

Faded's The Hive is a comprehensive zombie AI overhaul for Project Zomboid that transforms isolated zombie encounters into coordinated swarm behavior. Instead of every zombie acting independently, The Hive treats the zombie population as a single distributed organism that communicates through pheromone-like 'pings,' organizes around alpha leaders, and executes structured attacks including multi-directional raids, night hordes, and fence-climbing assault tactics.

Core Design Goals

- Zombies communicate indirectly - no telepathy, but spatial pheromone buckets spread target information.
- Threat scales with time - survival escalation, cataclysm nights, and bleed-tracking make longer survivors face harder odds.
- Server authoritative - all logic runs on the server; clients are passive viewers.
- Configurable - every mechanic can be enabled/disabled and tuned via sandbox settings.
- Performance-conscious - heavy computation is strided across ticks to avoid server spikes.

2. Architecture - How The Hive Is Built

File Structure

File	Side	Purpose
HiveMind.lua	Server	Shared mind layer - spatial buckets, pheromone pings, morning roost
HiveBehavior.lua	Server	Per-zombie AI engine - kind, light fear, flanking, silent strike
HiveAlpha.lua	Server	Alpha leader system - howls, echo cascades, bleed memory
HiveRaid.lua	Server	Coordinated raid director - sector assignment, multi-direction attacks
HiveNightHordes.lua	Server	Transient road horde spawner - scouts, rage cascades
HiveClimb.lua	Server	Fence/wall climb system - animations, stumble, topple
HiveMutations.lua	Server	Mutation trait system - evolved abilities on long-lived zombies
HiveReanimate.lua	Server	Corpse reanimation - delayed resurrection, modData tracking
HivePopulation.lua	Server	Density-based spawner - tier classification, escrow bank
HiveDarkNight.lua	Server	Dark night events - fog, reduced vision, climate effects
HiveNightCall.lua	Server	Player sound emission - draws indoor zombies toward players
HiveTactics.lua	Server	Feral state tracker - rage timers, building hostility marks
HiveConfig.lua	Shared	Sandbox var reader - centralised config access
HiveUtils.lua	Shared	Utility functions - spatial math, bucket helpers
HiveCompat.lua	Shared	Compatibility layer - cross-mod integration
HiveClient.lua	Client	Intentionally empty stub - mod is 100% server-side



Execution Model

The Hive uses a strided tick model. The game runs at 60 ticks/second but no single system runs every tick. Instead, each subsystem is scheduled to run every N ticks, spreading CPU load across frames. The main entry point is `Events.OnTick`, which acts as an orchestrator, forwarding control to each subsystem on its scheduled stride. Per-zombie logic fires through `Events.OnZombieUpdate`, called once per zombie per tick. All per-zombie work is guarded by a stride counter so each zombie only actually processes every few ticks, keeping overhead manageable even with hundreds of zombies.

Key Events Hooked

<code>Events.OnGameStart</code>	Single-player startup - initialises Hive state
<code>Events.OnServerStarted</code>	Dedicated server startup - initialises Hive state
<code>Events.OnTick</code>	Main orchestration loop (60/sec, strided per system)

<code>Events.OnZombieUpdate</code>	Per-zombie behavior engine (strided per zombie)
<code>Events.OnZombieDead</code>	Reanimate tracking + alpha death bookkeeping
<code>Events.OnHitZombie</code>	Climb system - hit feedback during fence crossing
<code>Events.OnAIStateChange</code>	Topple mechanic - fence climb failure knockdown
<code>Events.OnDeadBodySpawn</code>	Reanimate body registration
<code>Events.EveryOneMinute</code>	Stumble pass, alpha re-howl, toxicity tick

3. Core System: Shared Mind (Pheromone Layer)

`HiveMind.lua` implements the distributed intelligence of the Hive. Rather than every zombie independently detecting players, information spreads through a spatial bucket system that mimics pheromone communication.

Spatial Buckets

The map is divided into 24-tile buckets. Every active Hive zombie is assigned to the bucket containing its current position. When a zombie acquires a player target, it writes a 'ping' (a timestamped target coordinate) into its bucket.

Pings (Pheromones)

- When a zombie detects a player it deposits a ping into its spatial bucket.
- Ping TTL (time to live): 3,600 ticks » 60 seconds under normal conditions.
- Bleeding player TTL: 7,200 ticks » 2 minutes (alphas extend this).
- Other Hive members entering the same bucket inherit the ping's target on their next update.



- This creates a cascading awareness effect - one zombie alerting dozens nearby without any direct line-of-sight sharing.

Cell Population & Consensus Drift

Each bucket tracks a rolling count of Hive members present. During daytime, dormant zombies preferentially drift toward buckets with higher population counts. This causes natural clustering - large gatherings form at beacons where many zombies are already roosting, creating the impression of zombie congregations at key locations.

Morning Roost

During a 60-minute window centered on NightEndHour (dawn), special roost behavior fires:

- Outdoor zombies sprint toward their committed beacon locations.
- Indoor zombies transition to a docile state.
- The net effect: the swarm piles together at dawn rather than scattering, creating dangerous dawn clusters.
- Players who assume the night is over and step outside may encounter a dense, freshly roused mass.

4. Core System: Per-Zombie Behavior

HiveBehavior.lua runs on every Hive zombie every few ticks. At init, each zombie receives a permanent set of traits that determine how it behaves throughout its existence.

Zombie Kinds (Speed)

Setting	Default	Description
Shambler	~50%	Default slow walk speed. Remainder after fast/sprinter allocation.
Fast	35%*	Runs - faster than shambler, slower than sprinter. FastPercent setting.
Sprinter	15%*	Full sprint speed. SprinterPercent setting.

[OPERATOR NOTE] * Percentages are defaults. Adjust with SprinterPercent and FastPercent sandbox settings.

Light Fear

- 70% of Hive zombies (LightFearChance) are assigned fear of light.
- Fear intensity is rolled as 3-5 levels, affecting how strongly light repels them.
- When LightAmnesiaEnabled is true: bright light (above LightAmnesiaThreshold 0.78) causes zombies within LightAmnesiaNearRange (8 tiles) to forget their current target after LightAmnesiaMemorySec (5 seconds).
- Practical effect: torches, headlights, and generators can break zombie focus - but only on light-fearing ones. Flanking Each zombie rolls a flank channel: an angular offset of 0-5 from the player's facing direction. When a group converges on a player, they spread out in a fan rather than blob-rushing from one direction. This makes defensive positioning significantly harder.



Silent Strike

Within SilentStrikeRange (default 4 tiles), zombies lock onto a target using `setTarget()` and suppress world sounds. This prevents a zombie from alerting other nearby zombies when it closes in for the kill - the attack happens in near-silence.

Day Threat

When DayThreatEnabled is true, roosting zombies wake if players get within DayWakeRange (6 tiles) of the building they're roosting in. Players cannot safely walk past zombie-occupied buildings during the day - clearing is required.

Deceiver Zombies

DeceiverPercent (default 8%) of zombies appear docile and inactive, but attack when a player approaches within melee range. They look like regular dead or sleeping zombies until triggered.

[OPERATOR NOTE] Deceivers are particularly dangerous when looting buildings. A zombie that appears to be on the floor may actually be waiting.

Vision & Hearing Tiers

Each zombie rolls 1-3 perception tiers at init. Higher tiers increase night chase persistence - high-perception zombies track a fleeing player for longer in darkness.

5. Core System: Alpha Leaders

HiveAlpha.lua implements the Hive's leadership layer. Approximately AlphaPercent (10%) of zombies are designated alphas. Alphas prefer locations with low light, lead target acquisition, and amplify threat through sound cascades.

Howl Mechanics

- When an alpha acquires a player target, it emits a howl - a vanilla world sound that radiates AlphaCallRadius (40 tiles).
- The howl also deposits pings across the same radius, instantly alerting all Hive members in range.
- Alphas do NOT howl during daylight hours.
- Howls only fire at night (between NightStartHour and NightEndHour).

Bleed Escalation

- If a player is bleeding, the alpha's howl radius scales up by AlphaBleedMult (default 150 = $\times 1.5$).
- Bleeding players are tagged in modData - pings referencing them have extended TTL (2 minutes vs 60 seconds).
- Result: a single cut dramatically expands how far the Hive can track you.

Echo Cascade

Howls that reach other alphas trigger their own howls. The cascade propagates up to 3 hops. Each hop reduces the radius by $0.75\times$ (decay). A single initial howl can chain across a very large area:



Setting	Default	Description
Hop 0 (initial)	40 tiles	Original alpha howl - full radius
Hop 1	30 tiles	Second alpha - 40×0.75
Hop 2	22.5 tiles	Third alpha - 30×0.75
Hop 3	~17 tiles	Final hop - cascade ends

Ash Trail

When AlphaAshEnabled is true, alphas periodically (every AlphaAshIntervalSec = 30 seconds) emit an ash/particle effect in a AlphaAshRadius (2 tile) area around their spawn point. This is a visual indicator - experienced players can learn to spot alpha spawn zones.

6. Core System: Raid System

HiveRaid.lua implements the most dangerous night mechanic: coordinated multi-directional assault on player positions. The Raid Director runs approximately every 100 seconds at night.

How Raids Work

Step-by-step raid execution:

- Step 1: Director scans for alpha-bearing beacons within RaidRecruitTiles (200 tiles) of each online player.
- Step 2: Director selects 2-4 qualifying beacons.
- Step 3: Each beacon is assigned a compass sector: North, East, South, or West.
- Step 4: Director broadcasts a forced howl per sector, encoding the sector direction into the pings (RAID_N, RAID_E, RAID_S, RAID_W).
- Step 5: Each Hive member reads its assigned sector and approaches from that side.
- Step 6: Result: zombies close in from multiple directions simultaneously - no single 'safe' retreat direction.

Raid Parameters

Setting	Default	Description
RaidEnabled	true	Master toggle for the raid system.
RaidsPerNight	2	Number of raids that fire per night. Increase for harder servers.
RaidRecruitTiles	200	Tile radius searched for alpha beacons to assign sectors.
RaidDurationMin	6	Minimum raid duration in minutes before director considers it over.
RaidFeralBoost	25	Feral state boost percentage applied to raiders during a raid.



Cataclysm Gate

When SurvivalEscalation is enabled and players have survived 7 or more days, a Cataclysm Gate activates: raids per night are doubled (×2) and horde counts increase. This creates escalating pressure - the longer players survive, the more coordinated the Hive becomes.

[OPERATOR NOTE] Servers hosting long-term survivors (7+ in-game days) should account for Cataclysm escalation when setting raid counts. 2 raids/night becomes 4 automatically after day 7.

7. Core System: Night Hordes

HiveNightHordes.lua spawns transient, mobile zombie groups that patrol near players during the night phase. Unlike the static beacon population, hordes are created fresh each night and move.

Horde Spawning

- NightHordesPerNight (3) hordes spawn per night, one at a time with spacing.
- Each horde contains NightHordeMinSize-NightHordeMaxSize (8-16) zombies.
- Hordes spawn NightHordeMinDist-NightHordeMaxDist (60-120 tiles) from the nearest player.
- 15% of horde members (ScoutPercent) are designated scouts - they patrol in a wide arc around their spawn rather than charging.

Rage Cascade

When a horde member is wounded, there is a RageCascadeChance (30%) that nearby horde members enter a rage state. This can cascade - a single wounded zombie may trigger chain-aggression throughout the horde, causing them to sprint rather than shamle.

8. Core System: Climb System

HiveClimb.lua gives a portion of Hive zombies the ability to climb fences and walls, eliminating one of the most reliable player defenses.

Who Can Climb

- ClimbChance (40%) of all Hive zombies are capable climbers.
- Climbing only activates during raids or when the zombie is in a feral rage state.
- Passive zombies will NOT attempt to climb outside of these conditions.

Climb Animations

The system uses three custom animations:

Setting	Default	Description
HiveClimbWallStart	-	Animation that plays as zombie approaches and grips the fence.
HiveClimbWallSuccess	-	Animation for successful vault over the obstacle.



Setting	Default	Description
HiveClimbWallFail	-	Animation for failed attempt - zombie stumbles back.

Stumble & Topple

Even successful climbers can stumble:

- Sprinters and fast zombies: 5% stumble chance when vaulting.
- Shamblers: 2% stumble chance.
- When ToppleEnabled is true, a failed vault causes the zombie to be knocked down (topple state).
- This adds unpredictability - a raid may have one climber stumble, briefly blocking the path for others.

[OPERATOR NOTE] Fence perimeters are NOT safe during raids. Zombies will attempt to climb. Layered defenses (fence + wall + interior obstacles) are required to reliably stop climbers.

9. Core System: Mutation System

HiveMutations.lua represents evolved traits acquired by long-lived Hive members. Mutations increase the inherent danger of zombies that have existed for many in-game hours.

Mutation Chance

Setting	Default	Description
MutationEnabled	true	Master toggle for the mutation system.
MutationPercent	8	Base % chance a zombie has a mutation at init.
MutationNightBoost	10	Additional % added to mutation chance during night phase.

[OPERATOR NOTE] At night, effective mutation chance = MutationPercent + MutationNightBoost (8% + 10% = 18% effective).

10. Core System: Reanimate System

HiveReanimate.lua ensures that killing a zombie is not necessarily permanent. Corpses have a chance to rise again after a delay.

How Reanimation Works

- When a zombie dies, HiveReanimate rolls ReanimateChance (25%) to flag the corpse.
- Flagged corpses record a resurrection timestamp: a random delay between ReanimateMinHours and ReanimateMaxHours (12-36 in-game hours).
- The resurrection timestamp is stored in the corpse's modData - it survives save/load cycles.
- When the server clock passes the timestamp, the corpse stands up as a new Hive zombie.
- The newly risen zombie inherits a fresh Hive trait roll (kind, light fear, etc.).



[OPERATOR NOTE] Clearing an area does not guarantee safety. Corpses left behind have a 1-in-4 chance of rising within 12-36 hours. Burn or bag bodies to prevent reanimation.

Setting	Default	Description
ReanimateEnabled	true	Master toggle for corpse reanimation.
ReanimateChance	25	Percent chance any corpse will reanimate.
ReanimateMinHours	12	Minimum in-game hours before reanimation.
ReanimateMaxHours	36	Maximum in-game hours before reanimation.

11. Core System: Population & Repopulation

HivePopulation.lua manages how and where zombies spawn into the world. It scans the rooms around each player, classifies them by type, and queues spawn tasks at appropriate densities. Work is spread across many ticks to avoid server lag spikes.

Location Tiers & Density

Setting	Default	Description
commercial	18%	Shops, malls, retail buildings.
industrial	8%	Factories, warehouses, construction sites.
office	15%	Office buildings, corporate spaces.
residential	15%	Houses, apartments, suburbs.
service	15%	Gas stations, restaurants, service businesses.
boss	25%	High-value/high-danger locations (police, hospitals, military).

Escrow Bank

The population system uses an escrow model: spawn 'credits' accumulate over time and are spent when spawning zombies. This prevents sudden population bursts and smooths out density over time. AllowDebt (true) permits the system to spawn slightly ahead of credits when population is critically low.

Setting	Default	Description
HivePopulation	true	Master toggle for Hive population management.
ReserveInsidePct	40	% of spawn budget reserved for indoor locations.
FeralRoostPct	25	% of roosting zombies marked as feral (always aggressive).
AllowDebt	true	Allow spawn debt - spawn ahead of escrow when population is low.



12. Core System: Dark Night

HiveDarkNight.lua implements randomly occurring 'dark nights' - special nights where environmental conditions dramatically reduce visibility. Dark nights increase danger by making it harder to see approaching zombies.

When Dark Nights Occur

- Dark nights cannot occur until DarkNightMinDay (3) in-game days have passed.
- Each subsequent night rolls DarkNightChance (8%) for a dark night to trigger.
- Expected frequency: approximately 1 in 12.5 nights after day 3. Effects
- Fog intensity set to DarkNightFog (85) - heavy fog across the entire map.
- View distance reduced to DarkNightViewDistance (38 tiles).
- DarkNightClimate (true): weather effects applied alongside fog.
- Light threshold override via DarkNightLightThreshold (0.40) - artificial lights are less effective.
- DarkNightBucketPulseSec (30): darkness surges every 30 seconds, re-applying the effect.

[OPERATOR NOTE] Dark nights stack dangerously with Night Hordes and Raids - players caught outside during a dark night raid face significantly reduced reaction time.

13. Core System: Night Call

HiveNightCall.lua implements a passive threat mechanic: every online player periodically emits a world sound from their location during the night phase. This sound is heard by nearby zombies (including those inside buildings) and draws them toward the player. Mechanics

- Only fires during the Hive night window (NightStartHour to NightEndHour).
- Each player emits a sound every NightCallCooldown (5) seconds.
- Sound radius: NightCallRadius (1000 tiles) - effectively the entire loaded map.
- Sound volume: NightCallVolume (1000) - very high, waking indoor zombies.
- Effect: zombies that would otherwise stay inside buildings leave and stream toward player positions.

[OPERATOR NOTE] Night Call makes staying stationary indoors during the night increasingly dangerous. Zombies are drawn to player positions every 5 seconds regardless of what the player does. Barricading is essential - even a silent player attracts attention.

14. Core System: Tactics & Feral Marking

HiveTactics.lua tracks rage timers and feral state on a per-bucket basis. When a bucket (24-tile area) becomes actively hostile, it can be marked as 'feral' - all zombies within that bucket enter a permanent aggressive state until the building is cleared.

Feral State Triggers

- A raid entering a bucket marks it feral for the duration.



- A player killing zombies in a bucket without leaving may escalate the feral rating.
- FeralRoostPct (25%) of roosting zombies begin as feral regardless.
- Feral zombies do not respond to light fear or day docility - they remain aggressive always.

15. Core System: Toxicity

The toxicity system punishes players who kill too many zombies too quickly. When a player reaches the kill threshold, a toxic gas event is triggered at their position.

Toxicity Parameters

Setting	Default	Description
ToxicityEnabled	true	Master toggle for toxicity system.
ToxicityLeaderKills	5	Kills within the window that trigger a toxicity event.
ToxicityWindowMin	10	Tracking window in minutes - kills outside window don't count.
ToxicityRadius	45	Tile radius of the main toxic gas cloud.
ToxicityDurationHours	6	How long the gas persists (in-game hours).
ToxicityDamage	100	Damage per tick while in the gas cloud.
ToxicitySmallRadius	12	Radius for smaller secondary toxicity events.
ToxicitySmallDurationMin	45	Duration in minutes for small events.
ToxicitySmallDamage	35	Damage per tick for small events.

[OPERATOR NOTE] Toxicity discourages mass-clearing strategies. Players who try to sweep areas quickly will face a 45-tile poison cloud that persists for 6 in-game hours.

16. Core System: Migration

The migration system moves zombies across the world over time, ensuring areas that players have cleared do not remain permanently empty. Zombies gradually drift from dense areas toward player-cleared zones.

Setting	Default	Description
MigrationEnabled	true	Master toggle for zombie migration.
MigrationRingTiles	360	Tile ring size for migration scan - how far zombies can drift.

17. Timing & Phase Reference

All times are in-game hours unless noted. Real time varies by server time multiplier.



Setting	Default	Description
Night Phase Start	21:00	NightStartHour - Hive activates (raids, hordes, call, alphas)
Night Phase End	05:00	NightEndHour - Hive deactivates
Morning Roost Window	60 min	Centered on NightEndHour - outdoor zombies sprint to beacons
Raid Interval	~100 sec	Director evaluation cadence (real seconds)
Raid Min Duration	6 min	RaidDurationMin (real minutes)
Horde Spawn	Per night	NightHordesPerNight spawns across the night phase
Night Call Pulse	5 sec	NightCallCooldown (real seconds) - sound emitted per player
Alpha Ash Pulse	30 sec	AlphaAshIntervalSec (real seconds)
Dark Night Pulse	30 sec	DarkNightBucketPulseSec - darkness re-applied
Ping TTL (normal)	60 sec	3,600 ticks - pheromone fade time
Ping TTL (bleeding)	2 min	7,200 ticks - extended for bleeding players
Reanimate Delay	12-36 hrs	In-game hours before corpse rises (min-max)
Dark Night Frequency	~1/12 nights	After day 3; DarkNightChance 8% per night
Cataclysm Gate	Day 7+	Raid/horde counts double after 7 in-game days (if escalation on)
Toxicity Window	10 min	ToxicityWindowMin - kill tracking duration (real minutes)

18. Complete Sandbox Settings Reference

All settings live in the Hive page of the sandbox options. They can be changed at any time and take effect on the next relevant system evaluation.

Time / Phase

Setting	Default	Description
NightStartHour	21.0	Hour (0-24) when Hive night phase begins.
NightEndHour	5.0	Hour (0-24) when Hive night phase ends.
StormCountsAsNight	true	Treat heavy darkness/storms as the Hive night phase.

Zombie Speed

Setting	Default	Description
SprinterPercent	15	% of Hive zombies that sprint. Remainder after fast allocation are shamblers.
FastPercent	35	% of Hive zombies that run fast.

Light Behavior



Setting	Default	Description
LightFearChance	70	% of zombies that fear light sources.
LightThreshold	0.30	Light level (0-1) that counts as 'lit' for fear purposes.
LightAmnesiaEnabled	true	Bright light causes nearby zombies to forget their target.
LightAmnesiaNearRange	8	Tile range for light amnesia effect.
LightAmnesiaThreshold	0.78	Light level required to trigger amnesia.
LightAmnesiaMemorySec	5	Seconds the target memory is suppressed.

Flanking & Combat

Setting	Default	Description
FlankRadius	8	Tile radius for flank angle distribution - larger = wider fan.
SilentStrikeRange	4	Tiles within which zombies suppress world sounds during attack.

Day Threat

Setting	Default	Description
DayThreatEnabled	true	Roosting zombies wake when players pass within range.
DayWakeRange	6	Tile range triggering day-threat wake.

Setting	Default	Description
DeceiverPercent	8	% of zombies that appear docile but attack on close approach.

Night Ambush

Setting	Default	Description
NightAmbushEnabled	true	Enable coordinated night ambush behavior.
AllowDebt	true	Allow population escrow debt - spawn ahead of credits when low.

Population

Setting	Default	Description
HivePopulation	true	Enable Hive-managed population spawning.
ReserveInsidePct	40	% of spawn budget reserved for indoor locations.
FeralRoostPct	25	% of roosting zombies always in feral/aggressive state.

Raids



Setting	Default	Description
RaidEnabled	true	Enable the raid director system.
RaidsPerNight	2	Raids fired per night (doubled by Cataclysm Gate after day 7).
RaidRecruitTiles	200	Radius searched for alpha beacons to assign raid sectors.
RaidDurationMin	6	Minimum raid duration (real minutes).
RaidFeralBoost	25	Feral state boost % applied to raiders during a raid.

Night Hordes

Setting	Default	Description
NightHordesEnabled	true	Enable transient night horde spawning.
NightHordesPerNight	3	Number of hordes spawned each night.
NightHordeMinSize	8	Minimum zombies per horde.
NightHordeMaxSize	16	Maximum zombies per horde.
NightHordeMinDist	60	Minimum tile distance from player for horde spawn.
NightHordeMaxDist	120	Maximum tile distance from player for horde spawn.
ScoutPercent	15	% of horde members designated as patrol scouts.
RageCascadeChance	30	% chance wounded member triggers nearby horde rage.

Climbing

Setting	Default	Description
ClimbEnabled	true	Enable zombie fence/wall climbing.
ClimbChance	40	% of Hive zombies that can climb (only during raids/feral).
StumbleEnabled	true	Enable post-vault stumble chance.
ToppleEnabled	true	Enable fence topple/knockdown on failed climb.

Alpha Leaders

Setting	Default	Description
AlphaPercent	10	% of zombies designated as alpha leaders.
AlphaCallRadius	40	Howl radius in tiles (extended by bleed multiplier).
AlphaBleedMult	150	Bleed multiplier for howl radius ($150 = \times 1.5$).
AlphaAshEnabled	true	Enable ash trail visual effect from alphas.
AlphaAshRadius	2	Tile radius of ash particle effect.
AlphaAshIntervalSec	30	Seconds between ash pulses.



Mutations

Setting	Default	Description
<code>MutationEnabled</code>	true	Enable the mutation trait system.
<code>MutationPercent</code>	8	Base % mutation chance at zombie init.
<code>MutationNightBoost</code>	10	Additional % mutation chance during night phase.

Reanimate

Setting	Default	Description
<code>ReanimateEnabled</code>	true	Enable corpse reanimation.
<code>ReanimateChance</code>	25	% chance a corpse will reanimate.
<code>ReanimateMinHours</code>	12	Minimum in-game hours before resurrection.
<code>ReanimateMaxHours</code>	36	Maximum in-game hours before resurrection.

Night Call

Setting	Default	Description
<code>NightCallEnabled</code>	true	Enable player position sound emission at night.
<code>NightCallRadius</code>	1000	Sound emission radius in tiles.
<code>NightCallVolume</code>	1000	Sound volume (engine-arbitrary scale).
<code>NightCallCooldown</code>	5	Seconds between emissions per player.

Dark Night

Setting	Default	Description
<code>DarkNightEnabled</code>	true	Enable random dark night events.
<code>DarkNightChance</code>	8	% chance per night (after <code>DarkNightMinDay</code>) of a dark night.
<code>DarkNightMinDay</code>	3	Minimum in-game day before dark nights can occur.
<code>DarkNightFog</code>	85	Fog intensity 0-100 during dark nights.
<code>DarkNightViewDistance</code>	38	Tile view distance during dark nights.
<code>DarkNightClimate</code>	true	Apply weather effects alongside fog.
<code>DarkNightLightThreshold</code>	0.40	Light level override - lights less effective.
<code>DarkNightBucketPulseSec</code>	30	Seconds between darkness surge re-applications.

Toxicity



Setting	Default	Description
ToxicityEnabled	true	Enable kill-triggered toxicity events.
ToxicityLeaderKills	5	Kills within window that trigger large toxicity event.
ToxicityWindowMin	10	Kill tracking window in real minutes.
ToxicityRadius	45	Tile radius of the main toxic gas cloud.
ToxicityDurationHours	6	In-game hours the gas persists.
ToxicityDamage	100	Damage per tick in the gas cloud.
ToxicitySmallRadius	12	Radius for small secondary events.
ToxicitySmallDurationMin	45	Duration (real min) for small events.
ToxicitySmallDamage	35	Damage per tick for small events.

Migration

Setting	Default	Description
MigrationEnabled	true	Enable zombie population migration across the world.
MigrationRingTiles	360	Tile radius for migration drift scan.

Survival Escalation

Setting	Default	Description
SurvivalEscalation	true	Scale difficulty after 7+ in-game days (doubles raids/hordes).
CataclysmNightEnabled	false	Enable special cataclysm event nights (extreme difficulty).

Debug & Performance

Setting	Default	Description
DebugLog	false	Enable verbose debug logging to console.

Setting	Default	Description
ShowHUD	false	Display debug HUD overlay (server-side).
RenderCapEnabled	false	Cap the number of zombies rendered simultaneously.
RenderCapCount	140	Max renderable zombies if cap is enabled (range: 30-500).

19. Recommended Server Configurations

The following profiles are recommended starting points. Adjust based on your player count, playstyle, and desired difficulty. All unlisted settings use defaults.



Profile A - Casual / New Players

Lower intensity. Good for players new to the mod or zombie survival games.

Setting	Default	Description
SprinterPercent	5	Reduce sprinter count significantly.
FastPercent	20	Fewer fast zombies.
RaidsPerNight	1	One raid per night.
NightHordesPerNight	2	Two hordes per night.
NightHordeMaxSize	10	Smaller hordes.
AlphaPercent	5	Fewer alphas, shorter howl chains.
ClimbChance	20	Fewer climbers.
ReanimateChance	10	Lower corpse revival rate.
DarkNightChance	4	Rare dark nights.
SurvivalEscalation	false	Disable difficulty escalation after day 7.
ToxicityEnabled	false	Disable toxicity for easier combat.
DeceiverPercent	3	Fewer deceiver zombies.

Profile B - Standard Survival (Default)

Default settings. Balanced challenge for experienced players.

[OPERATOR NOTE] Run all defaults. See Section 18 for the complete default list.

Profile C - Hardcore / Military Roleplay

Maximum threat. Intended for coordinated groups who want brutal zombie pressure.

Setting	Default	Description
SprinterPercent	30	Nearly one-third of zombies sprint.
FastPercent	45	Another 45% run fast. Only 25% shamble.
RaidsPerNight	3	Three raids per night ($\times 2 = 6$ with Cataclysm).
NightHordesPerNight	5	Five hordes per night.
NightHordeMaxSize	24	Large hordes.
NightHordeMinDist	40	Hordes spawn closer.
AlphaPercent	18	Many alphas - frequent howl chains.

Setting	Default	Description
AlphaCallRadius	60	Extended howl radius.



Setting	Default	Description
AlphaBleedMult	200	Double radius when players bleed.
ClimbChance	60	Most Hive zombies can climb.
ReanimateChance	40	40% corpse revival.
ReanimateMinHours	6	Faster revival - 6-24 hour window.
ReanimateMaxHours	24	
DarkNightChance	15	~1 dark night per week of nights.
MutationNightBoost	20	Higher nighttime mutation rate.
ToxicityLeaderKills	3	Toxicity triggered by killing only 3 zombies quickly.
SurvivalEscalation	true	Escalation enabled - day 7+ is brutal.
CataclysmNightEnabled	true	Enable cataclysm nights for maximum pressure.
DeceiverPercent	15	15% of zombies are deceivers.

Profile D - PvE Server, Large Player Count

Optimized for 20+ players on one server. Reduces per-player pressure while maintaining world threat. Performance tunings included.

Setting	Default	Description
RaidsPerNight	1	One raid per night - each player group may still get hit.
NightHordesPerNight	2	Fewer hordes to limit server load.
NightHordeMaxSize	12	Moderate horde size.
RaidRecruitTiles	150	Tighter recruit radius - raids more localized.
MigrationRingTiles	240	Smaller migration ring - less global computation.
RenderCapEnabled	true	Cap zombie rendering.
RenderCapCount	200	Allow up to 200 rendered zombies.
NightCallRadius	600	Reduce night call radius for less global zombie movement.
SurvivalEscalation	false	Disable escalation - players have varied survival days.
DarkNightChance	5	Slightly reduce dark night frequency.

General Server Operator Tips

- Test configuration changes on a local single-player game before pushing to production.
- The Raid system is the highest-impact mechanic - adjust RaidsPerNight first when tuning difficulty.
- Night Call radius 1000 tiles is intentionally punishing. Reduce to 400-600 for more breathing room.
- Reanimate + large maps = corpse cleanup required. Server rules should reflect this.
- Cataclysm nights should only be enabled for groups that have played The Hive before.



- Alpha howl chains at high AlphaPercent can be very disruptive. Keep AlphaPercent below 20 for most servers.
- Climbers (ClimbChance 40%) make fence-only bases untenable. Communicate this to players.
- Toxicity punishes glass-cannon fighters. Consider disabling for servers focused on vehicle combat.

20. Companion Mod: Faded Trade Network

Faded Trade Network (FTN) adds a player-driven economy to multiplayer servers. Players can open storefronts, list items for sale, browse shops, and transact using physical Trade ID cards. All balances and listings are server-authoritative.

Core Concepts

- Trade ID Cards: Each player receives a physical FadedTradeNetwork.TradeID item. This card IS the wallet - losing it means losing access to the balance.
- Storefronts: Players claim a building as their shop. Sellers must be inside to list items; buyers must be physically inside to purchase.
- Currency: Base.Money and Base.MoneyBundle items are deposited to add balance to a Trade ID.
- Server Authority: All balances, listings, and transactions are validated server-side. Client UI is display-only.

Transaction Flow

- Step 1: Vendor deposits money items to load their Trade ID balance.
- Step 2: Vendor stands inside claimed storefront, opens seller panel, lists item. Item removed from inventory immediately.
- Step 3: Buyer enters storefront physically, opens buyer panel, selects item.
- Step 4: Server validates buyer is inside storefront, checks balance, debits buyer, credits seller, spawns item into buyer inventory.
- Step 5: Vendor can cancel listings from inside storefront (item returned to inventory).

Radio Browsing System

Shop owners can set a radio frequency for their storefront. Players with a powered/tuned walkie-talkie (or near a powered world radio) can browse stock remotely. Browse is view-only - purchases still require physical presence.

Admin Controls

Setting	Default	Description
AdminSetHourlyIncome	-	Set a player's hourly passive income amount.
AdminSetBalance	-	Directly set a player's Trade ID balance.
AdminAdjustBalance.nce	-	Add or subtract from a player's balance.
AdminIssueCard	-	Issue a new Trade ID card to a player.
AdminAddValuableItem	-	Add an item to the server's valuable items list.
AdminRemoveValuableItem	-	Remove an item from the valuables list.



[OPERATOR NOTE] Trade ID cards are physical items. If a player loses their card, their balance is inaccessible without admin intervention via AdminIssueCard + AdminAdjustBalance.

21. Companion Mod: Faded Afflictions

Faded Afflictions adds a server-managed psychological horror layer. Admins can assign hidden afflictions to players that cause audio hallucinations, visual distortions, panic spikes, and phantom figures - all without any direct game threat. The goal is sustained psychological pressure.

Affliction Types

Setting	Default	Description
Paranoia	Sev 1-3	Fake footsteps, growls, screen shadows, shadow figures, whispers, darkness surges.
Phantom Sounds	Sev 1-3	Fake knocks/windows, footsteps, whispers, zombie audio, halo text overlays.
Shattered Mind	Sev 2-4	Panic/stress spikes, whispers, laughter, player shouts, shadows, fog hazes.
Cursed Fog	Variable	Persistent fog overlays and disorientation effects.
The Shroud	Advanced	Proximity-based audio escalation - worsens as zombies get closer.

Effect Types

Per-affliction effects that fire based on chance, cooldown, and context multipliers:

- Audio: fake_knock, fake_window, fake_footsteps, fake_whisper, whisper_zombie, fake_growl, fake_laughter, zombie_audio, halo_whisper, player_shout
- Emotional: panic_spike, stress_spike
- Visual: screen_shadow, shadow_figure, fog_haze, darkness_surge, cold_spot_haze

Context Multipliers

Effect chance increases based on player state:

Setting	Default	Description
Night time	+35%	All effects more likely during darkness.
Panic > 40	+20%	High panic amplifies effect rate.
Stress > 0.45	+20%	High stress amplifies effect rate.
Health < 65	+15%	Injured players experience more effects.

Cure System

- Stabilizing Pills: Reduces panic (-12) and stress (-0.10). Suppresses effects for StabilizingPillsSuppressionHours (8 hours).



- Admin Commands: Admins can directly remove any affliction via chat commands (AllowAdminCommands must be true).
- Server Events: Optional automatic cures triggered by in-game milestones.
- PersistsThroughDeath: Configurable per affliction type. If true, death does not clear the affliction.

Key Settings

Setting	Default	Description
EnableMod	true	Master toggle for affliction system.
AllowAdminCommands	true	Admins can apply/remove afflictions via chat.
PersistThroughDeathDefault	true	Afflictions survive player death by default.
UseSteamID	true	Use Steam ID for identity (vs username on offline servers).
EnableVisualHallucinations	true	Enable visual symptom effects.
EnableHaloText	true	Enable halo text whisper overlays.
EnableShadowFigureHallucinations	true	Enable SCP-style phantom figures.
EnableFakeShadowZombies	false	Enable extra dangerous phantom zombie flashes.
MaxEffectsPerHour	4	Cap on triggered effects per hour per player.
CureItemsEnabled	true	Allow Stabilizing Pills as a cure.
StabilizingPillsSuppressionHours	8	Hours pills suppress effects.

22. Companion Mod: CSR Claims System

Common Sense Reborn - Claims is a server-authoritative safehouse and building claim system. Players claim buildings as personal or faction property, set role-based permissions, invite members, and the server enforces access control.

Claim Registry

- Each claim is a flat ModData row: id, x, y, w, h, owner, kind (personal/faction), title, members, roles.
- On server start, existing vanilla SafeHouses are automatically migrated into the claim registry (one-time migration).
- The server maintains a Java SafeHouse mirror for loot respawn compatibility.
- Claims persist across save/load via server Global ModData.

Padlock System

- Players can padlock containers and vehicles with a physical Padlock item.
- The padlock creates a Key1 item with a unique 8-character hex hash stored in modData.
- Only players with a matching key item can open the padlocked container/vehicle.
- Bolt cutters can break padlocks (validated client-side, proximity + sandbox flag checked server-side).



- Original Padlock item is consumed on use.

Permission Roles

Setting	Default	Description
Owner	-	Full control - modify claim, invite members, transfer, set roles.
Admin	-	Can invite members, modify some settings, access all containers.
Member	-	Access containers and spaces within the claimed building.

Commands Dispatched by Server

Setting	Default	Description
CSR_ClaimAdded	Broadcast	New claim registered - sent to all clients.
CSR_ClaimRemoved	Broadcast	Claim released - sent to all clients.
CSR_ClaimUpdated	Broadcast	Claim row mutated - sent to all clients.
CSR_ClaimsBundle	Targeted	Full claim registry sent to connecting client.
CSR_ClaimResult	Targeted	Success/failure result sent to requesting player.

[OPERATOR NOTE] NEVER route player claims through `SafeHouse.addSafeHouse()` server-side. CSR uses only the bounds-overload form. This is a hard architectural rule.

23. Data Persistence & Save/Load

The Hive suite persists all critical state across server restarts and player reconnects.

Storage Locations

Hive zombie traits	zombie.modData - keys: Hive_native, Hive_kind, Hive_alpha, Hive_lightFear, Hive_t
Hive beacons	Disk save every ~100 seconds via Hive.Beacons.save()
Hive corpse reanimate	Corpse modData - resurrection timestamp stored on body object
FTN balances	Server Global ModData under FadedTradeNetwork key
FTN storefronts	Server Global ModData - claim bounds, listing data, radio frequencies
FTN trade IDs	Server Global ModData - identity records keyed by card ID
Afflictions data	SModData under FadedAfflictions_Data_v1 key
CSR claims	Server Global ModData - flat row format, all primitives
CSR padlocks	container/vehicle modData - csrPadlocked, csrPadlockClaim, csrPadlockKeyHash



Save/Load Safety

- All Hive zombie modData survives save/load - zombie traits and reanimate timestamps persist.
- Beacon data is explicitly saved every ~100 sec - server crashes between saves may lose up to 100 sec of beacon state.
- FTN and Afflictions use Global ModData / SModData which are saved with the world.
- CSR claim migration runs only once (MIGRATION_FLAG guards it) - vanilla safehouses are imported exactly once.

24. Performance Considerations

The Hive is designed to be server-friendly, but high zombie populations and aggressive settings will increase server load. Understanding the performance model helps server operators tune appropriately.

Tick Striding

No system runs every tick. Each subsystem runs every N ticks with N chosen to spread load. Per-zombie OnZombieUpdate also strides - each zombie only fully processes every few ticks. This means higher zombie counts have a roughly linear (not exponential) cost impact.

High-Cost Systems (in order)

Setting	Default	Description
Population Scanner	High	Scans rooms around every player each evaluation. Reduce by lowering player c
Migration System	Medium	Scans MigrationRingTiles radius - reduce MigrationRingTiles to lower cost.
Raid Director	Low	Runs every ~100 sec - minimal ongoing cost.
Night Hordes	Low	Spawns 3 hordes per night - small count, low cost.
Alpha Howl Cascade	Low	Only fires when alpha acquires target at night - event-driven.
Night Call	Low	Sound emit per player every 5 sec - very cheap.

Recommended Performance Tunings for Large Servers

- Enable RenderCapEnabled and set RenderCapCount to 150-200.
- Reduce MigrationRingTiles to 200-240.
- Reduce NightCallRadius to 500-700 (less global zombie movement).
- Reduce RaidRecruitTiles to 120-150 (more localized raids).
- Reduce NightHordesPerNight to 1-2.
- Increase Night Horde min spawn distance (NightHordeMinDist 80+) to prevent tight clustering near players.



25. Troubleshooting & Debug Tools

Debug Logging

Set `DebugLog = true` in sandbox settings to enable verbose logging to the server console. This logs bucket operations, ping deposits, alpha howls, raid assignments, horde spawns, and reanimate events. Disable on live servers - output volume is high.

Debug HUD

Set `ShowHUD = true` to enable a server-side debug overlay. Shows active Hive state: bucket populations, active raids, feral zone markers, current phase (night/day). Disable on production servers.

Common Issues

Setting	Default	Description
Raids not firing	Night phase	Verify <code>NightStartHour/NightEndHour</code> match server clock. Check <code>RaidEnabled = true</code> .
No alpha howls	Night only	Alphas only howl at night. If daytime, this is expected.
Zombies not climbing	Raid/feral	Climbing only activates during raids or feral state. Not passive behavior.
Corpses not reanimating	Timing	Minimum delay is 12 in-game hours. Check <code>ReanimateEnabled = true</code> .
Night Call too aggressive	Settings	Reduce <code>NightCallRadius</code> (1000 default is very large). Try 400-600.
Server lag during night phase	Performance	Enable <code>RenderCap</code> , reduce <code>MigrationRingTiles</code> , reduce horde counts.
Toxicity killing players unfairly	Settings	Increase <code>ToxicityLeaderKills</code> threshold or disable with <code>ToxicityEnabled = false</code> .
Dark nights too frequent	Settings	Reduce <code>DarkNightChance</code> from 8 to 3-5.
FTN balance lost	Trade ID card	Player lost their Trade ID item. Use <code>AdminIssueCard + AdminAdjustBalance</code> .
CSR claim not visible	Migration	Check <code>MIGRATION_FLAG</code> in <code>ModData</code> . If set twice, CSR may double-import safehouses.

Beacon Save Interval

Beacons are saved every ~100 server seconds. If the server crashes between saves, up to 100 seconds of beacon state may be lost. This is generally acceptable - beacons rebuild as zombies re-populate areas.

Compatibility Notes

- The Hive overrides vanilla zombie speed/behavior - disable other zombie speed mods to avoid conflicts.
- FTN and CSR Claims do not conflict. They use separate `ModData` namespaces.
- Faded Afflictions applies on top of The Hive - affliction effects layer with Hive night mechanics.
- CSR Claims and vanilla SafeHouses coexist via the migration system (one-time import).
- `HiveClient.lua` is intentionally empty - no client mod load required. Install server-side only.